CPEN 416

Lecture 16: Expectation values, mixed states, and density matrices

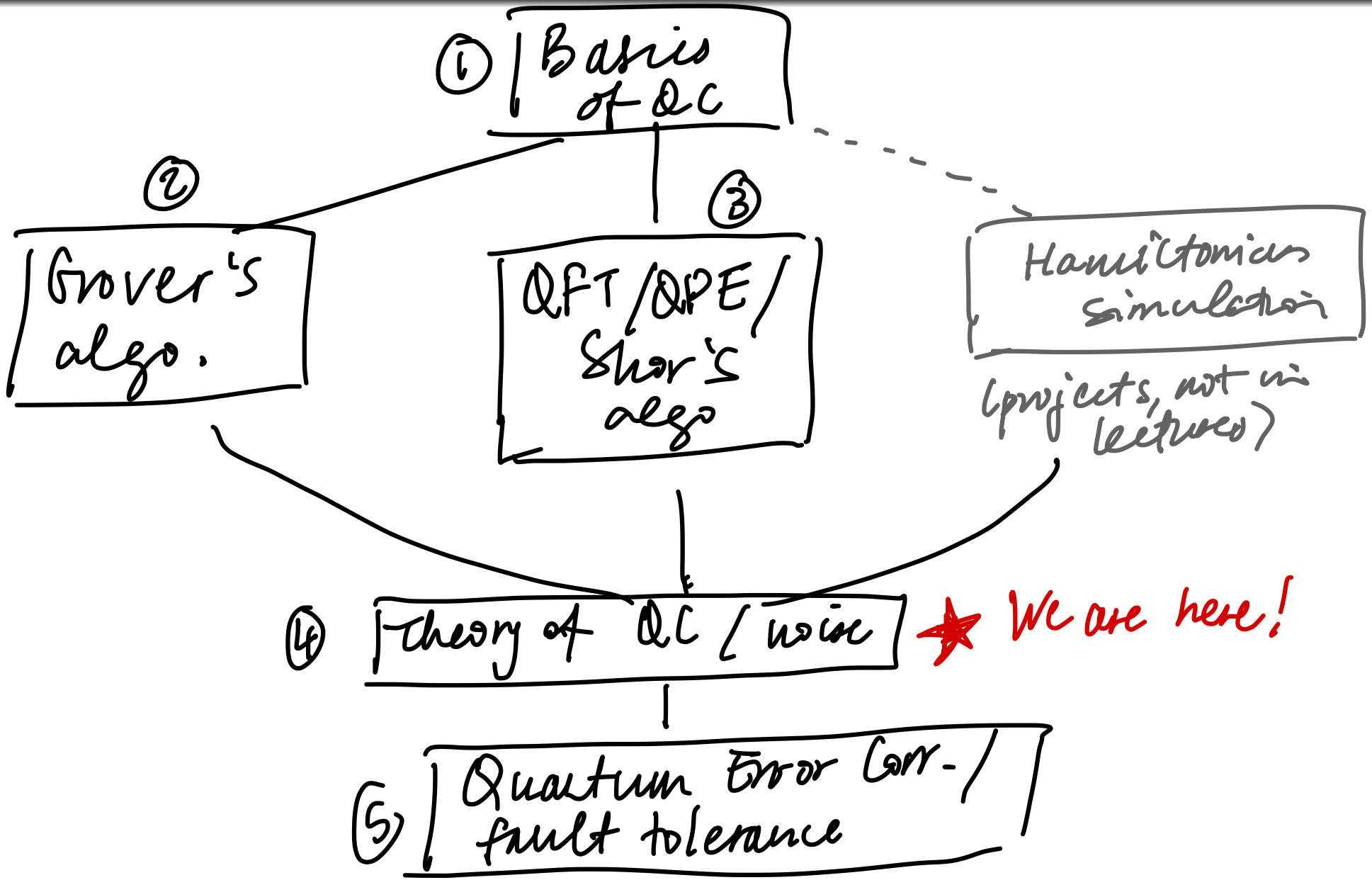
Thursday 5 November 2026

Announcements

fill out weekly surveys!

- Friday office hours cancelled due to travel
- Project grading rubric available today
- Reading break next week - no tutorial, no Tuesday class
  - Next class Thurs. 12 Nov.
  - Next quiz Tues. 17 Nov. (L16-L17)
- Bonus reading break assignment on Piazza

Where are we going?



Learning outcomes

- Define an observable and compute its expectation value
- Define a *mixed state*, and express quantum states as density matrices
- Identify mixed states on the Bloch sphere

Review: projective measurements

\varphi

\phi

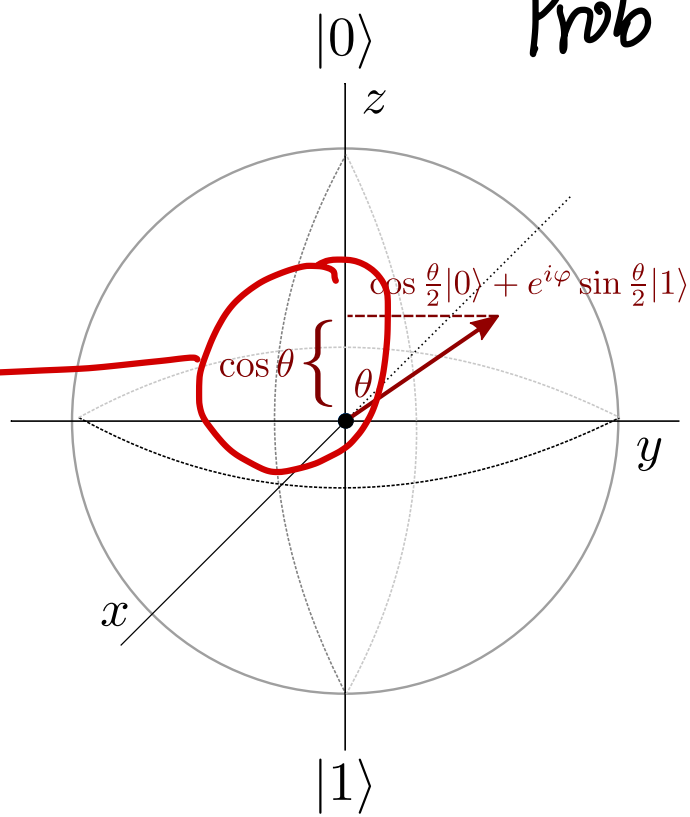
Recall: when we measure a qubit in state $|\varphi\rangle$ with respect to basis $\{|\psi_i\rangle\}$, the probability of observing it in state $|\psi_i\rangle$ (outcome i) is

$$\Pr(\text{outcome } i) = |\langle \psi_i | \varphi \rangle|^2$$

If we observe outcome i, the qubit *remains in* $|\psi_i\rangle$.

Review: projective measurements

Projective measurement corresponds to a geometric projection onto the axis represented by a basis.



Prob $|0\rangle = |\langle 0|\psi\rangle|^2$

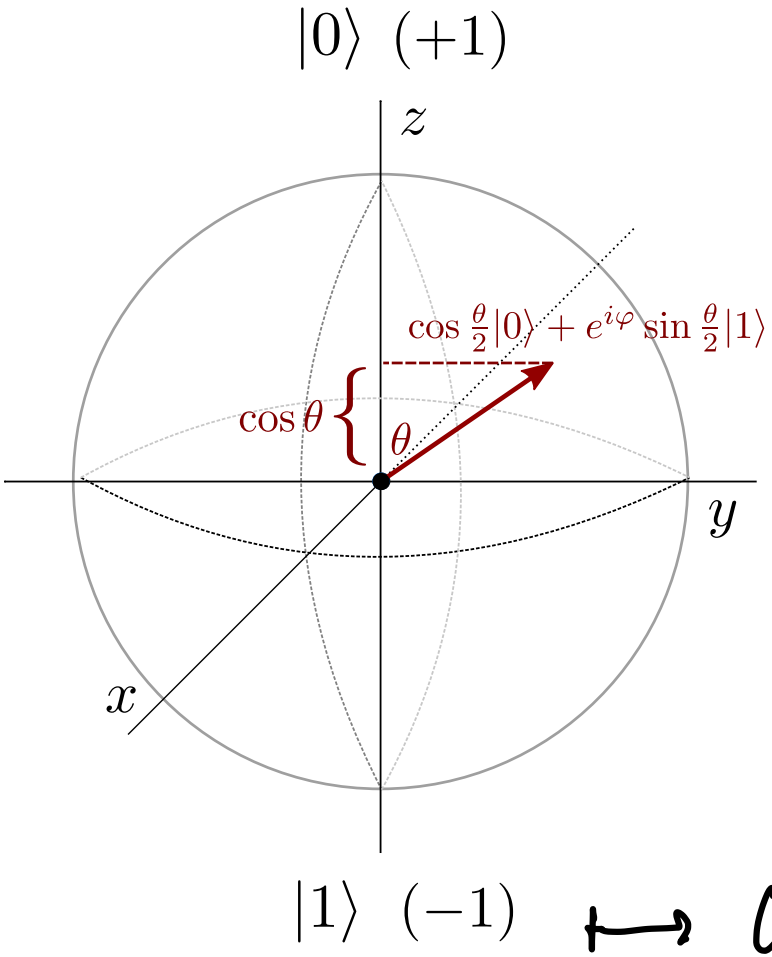
$= \cos^2\frac{\theta}{2}$

$= \frac{1+\cos\theta}{2}$

not prob. 0?, but expectation value of diff. kind of meas.

Let's look at measurement in a slightly different way...

Expectation values



Assign each basis states a real-valued outcome:

$|0\rangle \mapsto +1 \quad |1\rangle \mapsto -1$

If we prepare and measure N times, what is the **expected value** of the meas. outcome?

Saw $|0\rangle$ n_0 times
$|1\rangle$ n_1 times

(weighted average)

In the limit of infinite shots...

exp. val: $\frac{(1)\cdot n_0 + (-1)\cdot n_1}{N}$

$\longmapsto (1)\cdot p_0 + (-1)\cdot p_1 = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$

$= \cos\theta$

Expectation values

Spectral theorem: $M = \sum_i \lambda_i |v_i\rangle\langle v_i|$ (λ_i: eigval, $|v_i\rangle$: eigvec.)

This assignment is no accident...

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Its eigensystem is

$|v_0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$ $\lambda_0 = +1$ $\qquad |v_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$ $\lambda_1 = -1$

$$\langle Z \rangle = \lambda_0 \cdot |\langle 0|\psi\rangle|^2 + \lambda_1 \cdot |\langle 1|\psi\rangle|^2$$
$$= \lambda_0 \langle\psi|0\rangle\langle 0|\psi\rangle + \lambda_1 \langle\psi|1\rangle\langle 1|\psi\rangle$$
$$= \langle\psi| \left[\lambda_0 |0\rangle\langle 0| + \lambda_1 |1\rangle\langle 1|\right] |\psi\rangle$$

$= Z = 1 \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + (-1) \cdot \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$= \langle\psi|Z|\psi\rangle$$

Z is an example of a more general concept: an observable.

Observables

Generally, we are interested in measuring real, physical quantities. In physics, these are called observables.

Mathematically, an observable M is a Hermitian operator (matrix)

$$M = M^\dagger \qquad \begin{pmatrix} 3 & -4i^* \\ 4i & 3 \end{pmatrix} \checkmark$$

Analytically, the **expectation value** of measuring the observable M given the state $|\psi\rangle$ is

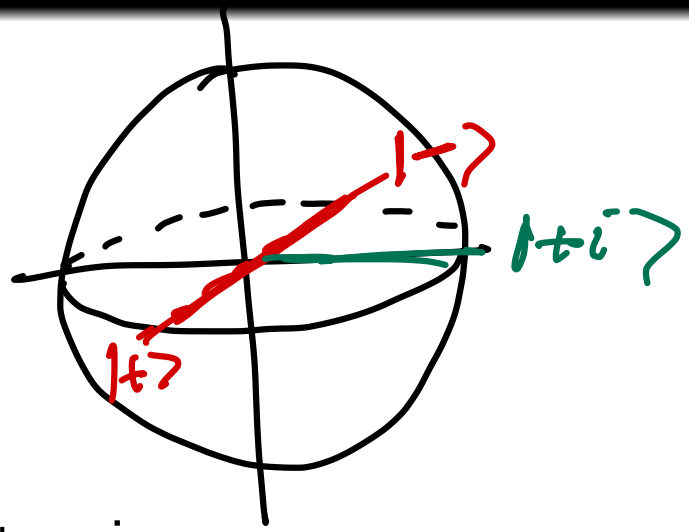
$$\langle M \rangle = \langle \psi | M | \psi \rangle$$

Why Hermitian? eigenvalues are REAL.

Observables

Example:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$



X is Hermitian and its (normalized) eigensystem is

$$|v_0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle \qquad \lambda_0 = +1$$

$$|v_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle \qquad \lambda_1 = -1$$

$$\langle X \rangle \text{ for } |+i\rangle = 0$$

Observables

Example:

$$Y|v_0\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Y is Hermitian and its (normalized) eigensystem is

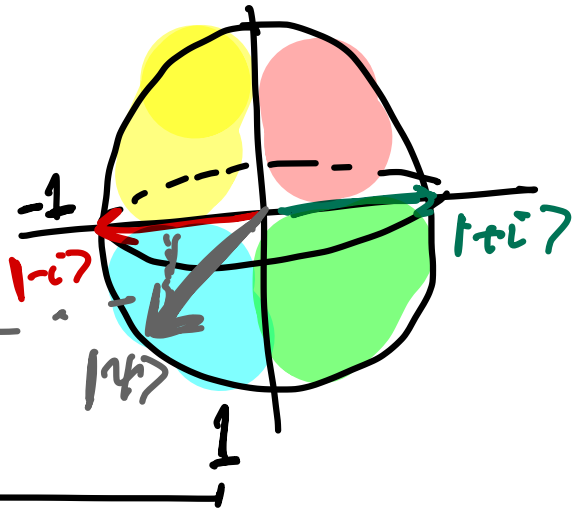
$$|v_0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = |+i\rangle \qquad \lambda_0 = +1$$

$$|v_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = |-i\rangle \qquad \lambda_1 = -1$$

Expectation values: analytical

Exercise: consider the quantum state

$$|\psi\rangle = \frac{1}{2}|0\rangle - i\frac{\sqrt{3}}{2}|1\rangle.$$



Compute the expectation value of Y:

Range of exp vals: -1 to 1

$$\langle\psi|Y|\psi\rangle = \left(\frac{1}{2}\langle 0| + i\frac{\sqrt{3}}{2}\langle 1|\right) Y \left(\frac{1}{2}|0\rangle - i\frac{\sqrt{3}}{2}|1\rangle\right)$$

$$= \left(\frac{1}{2}\langle 0| + i\frac{\sqrt{3}}{2}\langle 1|\right)\left(\frac{i}{2}|1\rangle - \frac{\sqrt{3}}{2}|0\rangle\right)$$

$$= \frac{i}{4}\langle 0|1\rangle - \frac{\sqrt{3}}{4}\langle 1|1\rangle - \frac{\sqrt{3}}{4}\langle 0|0\rangle - \frac{i3}{4}\langle 1|0\rangle$$

($\langle 1|1\rangle = 1$, $\langle 0|0\rangle = 1$)

$$= -\frac{\sqrt{3}}{2}$$

= Prob(State obs in $|+i\rangle$) − Prob(state obs in $|-i\rangle$)

Expectation values

Let's do this in PennyLane instead:

```
dev = qp.device('default.qubit', wires=1)

@qp.qnode(dev)
def measure_y():
    qp.RX(2*np.pi/3, wires=0)
    return qp.expval(qp.PauliY(0))
```

Multi-qubit expectation values

Example: operator $Z \otimes Z$.

$$H = -\sum J Z_i Z_j - \sum h X_i$$

Eigenvalues are computational basis states:

$$(Z \otimes Z)|00\rangle = |00\rangle$$
$$(Z \otimes Z)|01\rangle = -|01\rangle$$
$$(Z \otimes Z)|10\rangle = -|10\rangle$$
$$(Z \otimes Z)|11\rangle = |11\rangle$$

Total energy:
$+\langle X \rangle$ for each qubit
$+\langle ZZ \rangle$ for each qubit pair
(scaled by $-h$ and $-J$ resp.)

To compute an expectation value from data:

$$\langle Z \otimes Z \rangle = \frac{1 \cdot n_1 + (-1) \cdot n_{-1}}{N}$$

Multi-qubit expectation values

Example: operator $X \otimes I$.

Eigenvalues of X are the $|+\rangle$ and $|-\rangle$ states:

$$
\begin{aligned}
(X \otimes I)|+0\rangle &= |+0\rangle \\
(X \otimes I)|+1\rangle &= |+1\rangle \\
(X \otimes I)|-0\rangle &= -|-0\rangle \\
(X \otimes I)|-1\rangle &= -|-1\rangle
\end{aligned}
$$

Fun fact: All Pauli operators have an equal number of $+1$ *and* -1 *eigenvalues!*

Multi-qubit expectation values

How to compute expectation value of X from data, when we can only measure in the computational basis?

Basis rotation: apply H to first qubit

$$(H \otimes I)(X \otimes I)|+0\rangle = |00\rangle$$
$$(H \otimes I)(X \otimes I)|+1\rangle = |01\rangle$$
$$(H \otimes I)(X \otimes I)|-0\rangle = -|10\rangle$$
$$(H \otimes I)(X \otimes I)|-1\rangle = -|11\rangle$$

When we measure and obtain $|10\rangle$ or $|11\rangle$, we know those correspond to the -1 eigenstates of $X \otimes I$.

Hands-on: multi-qubit expectation values

Multi-qubit expectation values can be created using `@` :

```
@qp.qnode(dev)
def circuit(x):
    qp.Hadamard(wires=0)
    qp.CRX(x, wires=[0, 1])
    return qp.expval(qp.PauliZ(0) @ qp.PauliZ(1))
```

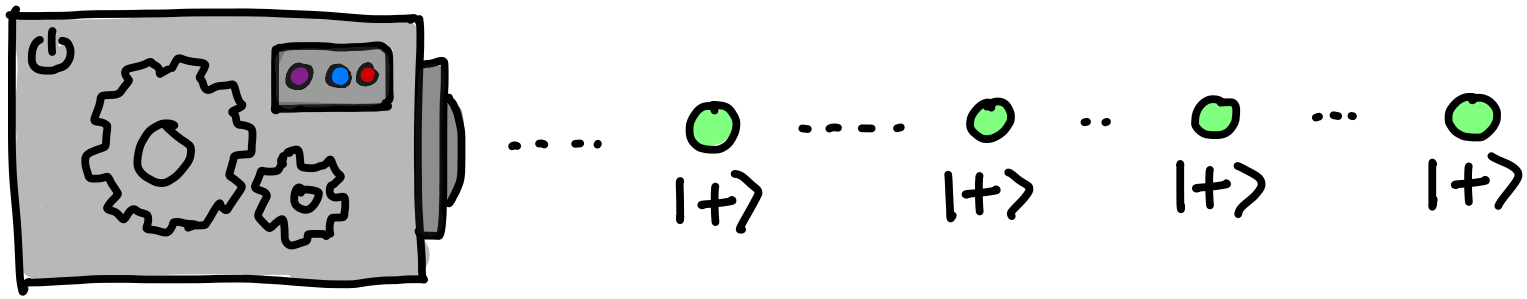
$U = e^{-iHt} \Rightarrow$ H Herm., U is unitary

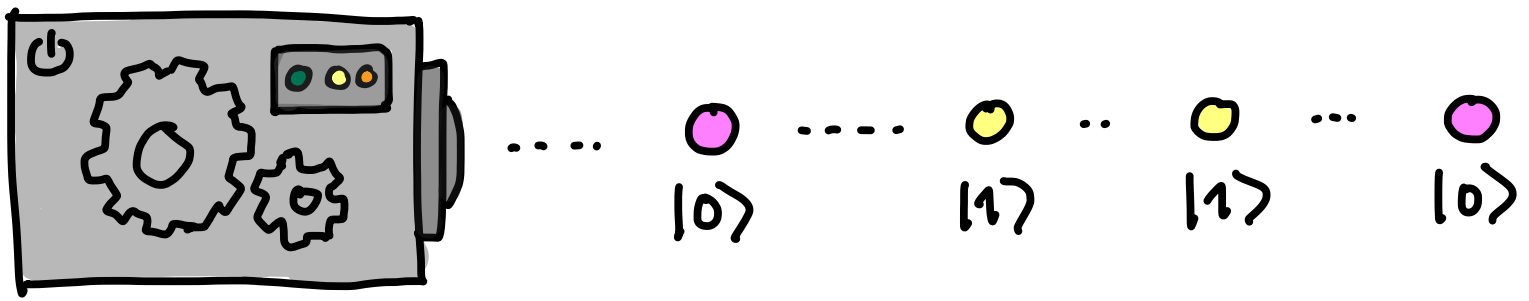
Can return *multiple* expectation values if no shared qubits.

```
@qp.qnode(dev)
def circuit(x):
    qp.Hadamard(wires=0)
    qp.CRX(x, wires=[0, 1])
    return qp.expval(qp.PauliZ(0)), qp.expval(qp.
                                    PauliZ(1))
```

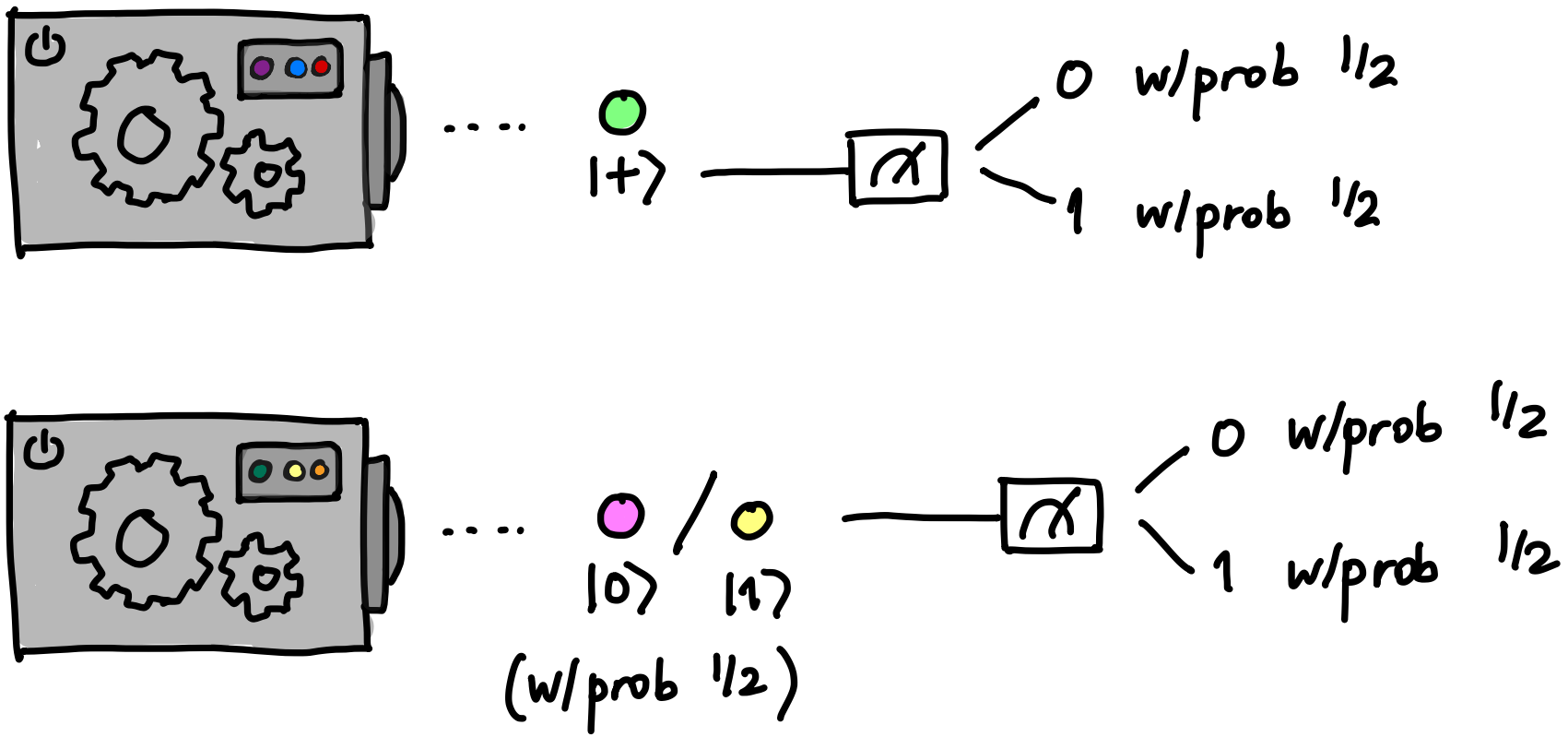
Mixed states

Are these two devices the same?

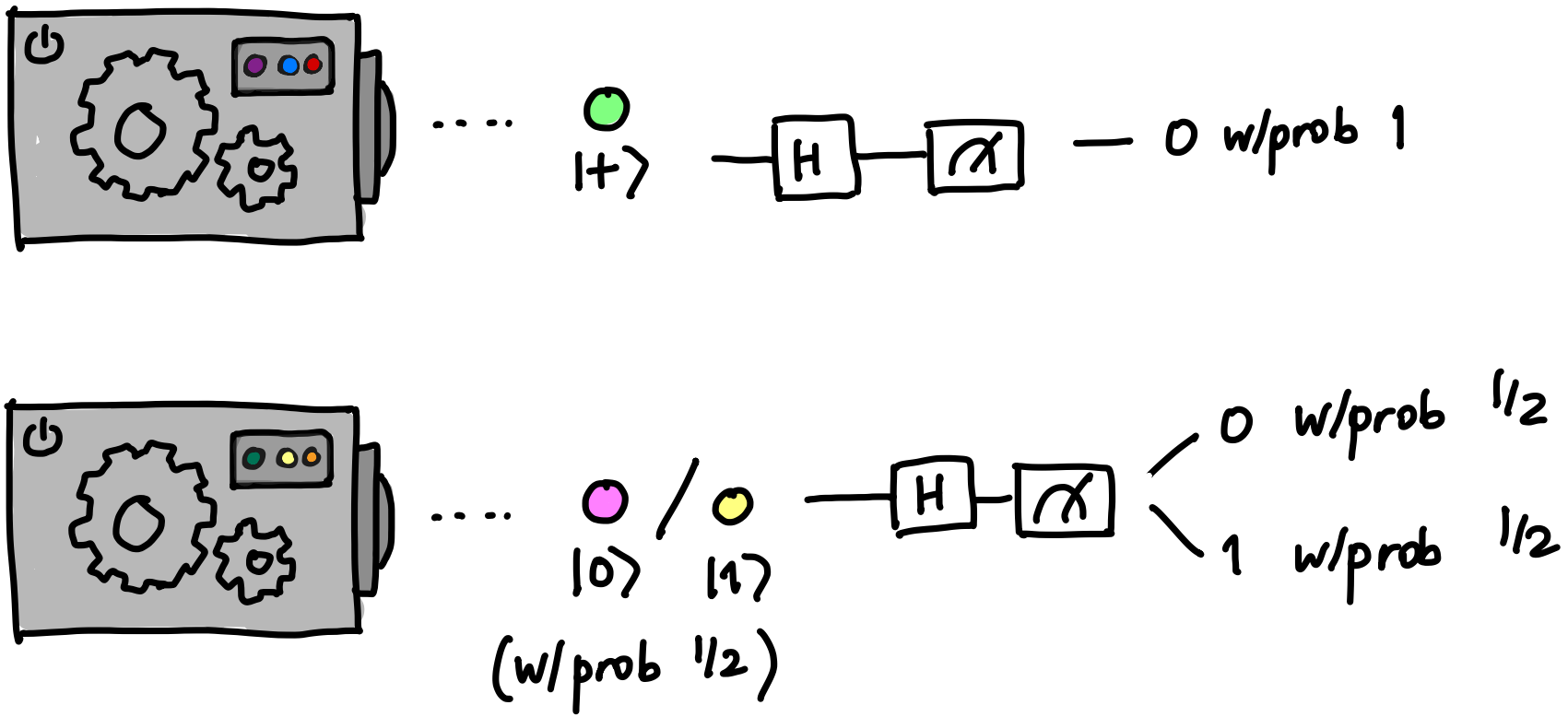




Mixed states



Mixed states



What is the second box doing?

Mixed states

The second box prepares a **mixed state**.

A **pure state** can be expressed as a single ket vector, e.g.,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

A **mixed state** is a *probabilistic mixture of pure states*. It is represented a **density matrix**.

Density matrices

The density matrix of a pure state $|\psi\rangle$ is

$$\rho = |\psi\rangle\langle\psi|$$

\rho

Exercise: what are the density matrices for $|0\rangle$ and $|1\rangle$?

$$\rho_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\rho_1 = |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Density matrices

Density matrices of mixed states are linear combinations of those of pure states:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \qquad \sum_i p_i = 1$$

Exercise: A system prepares $|+\rangle$ with probability 1/3, and $|0\rangle$ with probability 2/3. What is its state?

$$\rho = \frac{1}{3}|+\rangle\langle+| + \frac{2}{3}|0\rangle\langle 0|$$

$$= \frac{1}{3}\cdot\frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} + \frac{2}{3}\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}$$

$$= \begin{pmatrix}5/6 & 1/6\\ 1/6 & 1/6\end{pmatrix} \longrightarrow$$

properties: Hermitian
diag. sums to 1

Density matrices

Density matrices have some nice properties.

- they are Hermitian
- they have trace 1
- they are positive semi-definite (all eigenvalues are ≥ 0)
- (for pure states only) they are projectors, i.e., $\rho^2 = \rho$

Check with our example:

$\rho = |\psi\rangle\langle\psi|$

$\rho^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi|$

$= \rho$

Fun activity: show properties hold for general $\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i|$

Density matrices and expectation values

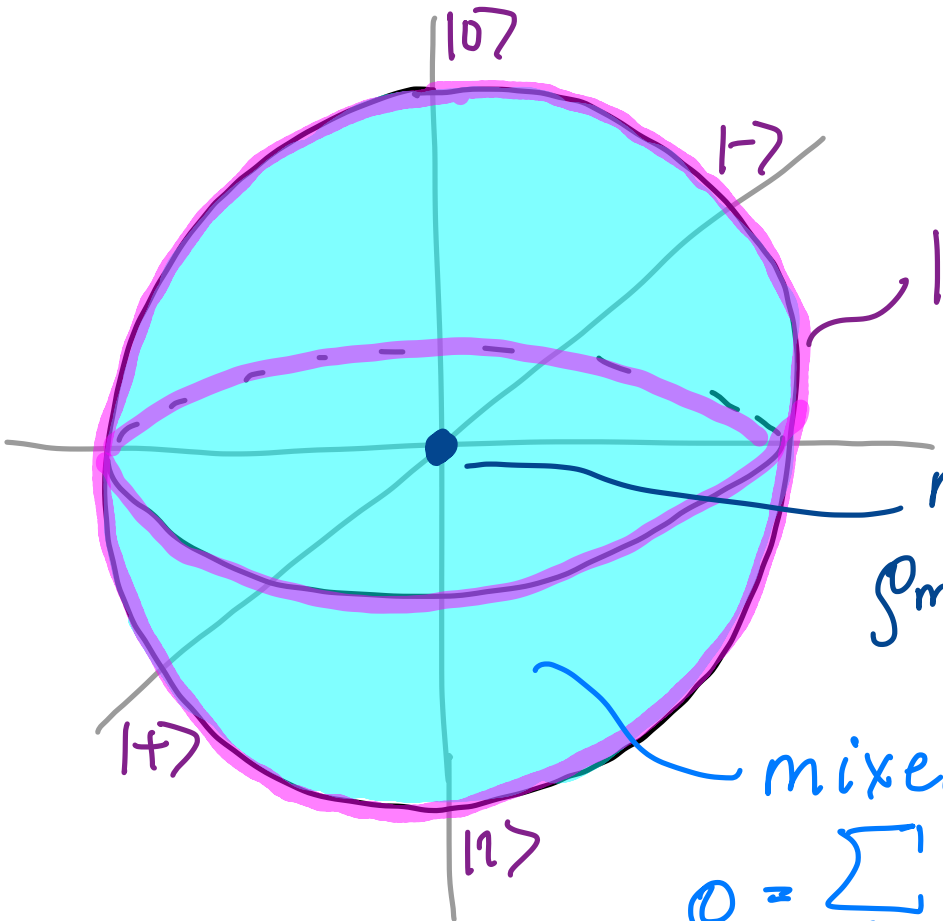
Exercise: A system prepares $|+\rangle$ with probability 1/3, and $|0\rangle$ with probability 2/3.

$$\rho = \begin{pmatrix} \frac{5}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} \end{pmatrix}$$

What are the expectation values $\langle X \rangle$,$\langle Y \rangle$, $\langle Z \rangle$?

☆ start here next class.

Density matrices and the Bloch sphere



pure states

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

maximally mixed state

$\rho_{max} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\mathbb{1}$

mixed states

$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$

Next time

Next class:

- Operations and measurements on mixed states
- Quantum channels

Action items:

1. Work on project

Recommended reading:

- Codebook module NT
- Nielsen & Chuang 2.4, 2.2.5-2.2.6